

Swan Teleco Retention Analysis

Customer churn, demand drivers, and a targeted retention plan

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The Brief & Our Approach

What was asked, and how we answered it

The retention team requested a comprehensive analysis to identify customer churn, understand contributing factors, and deliver actionable insights to enhance retention strategies and improve customer satisfaction.

Who is churning?

We investigated demographics, tenure, and the products churners hold.

What drives it?

Which factors raise or lower churn odds?

Who to mail?

The 500 retained customers most likely to churn.

What's the risk?

A churn-risk score for every remaining customer.

Our approach: clean and validate the SCV → engineer and encode features → fit and evaluate a logistic regression churn model → translate the predictions into a mailer list, a full risk register, and a set of recommendations.

The Scale of the Problem

Cleaning, Processing, and Modelling

1,869

of 7,043 customers left last quarter – a 26.5% churn rate

Clean

11 blank Total Charges values (new joiners) were found and fixed, standardised “No service” labels to “No” and validated the full pipeline with automated checks.

Engineer

Dropped identifiers and leakage columns, one-hot encoded 17 customer attributes, and standardised tenure.

Model

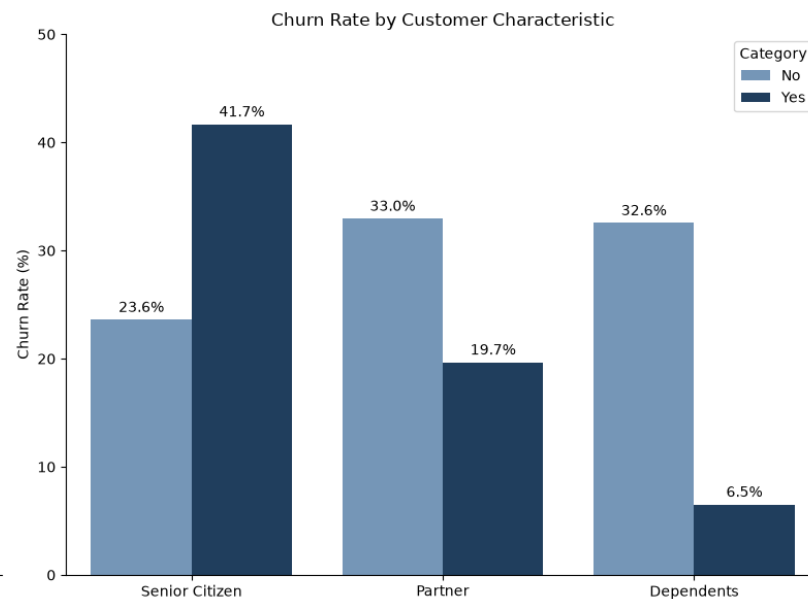
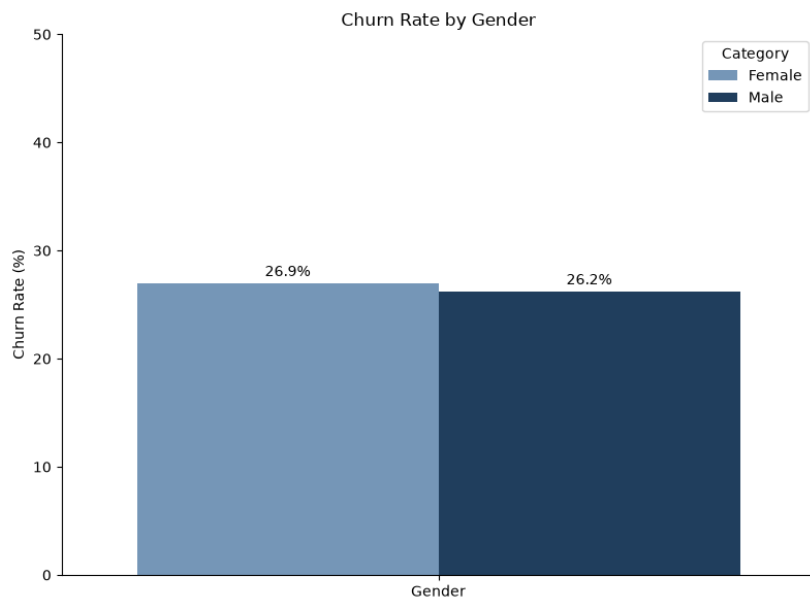
An 80/20 stratified train-test split feeding a logistic regression classifier, so every driver is directly interpretable as a churn odds ratio.

Evaluate

80% test accuracy and 0.85 AUC, with the classification threshold tuned for the recommendations that follow.

Who Is Churning

Demographics



Household status is what matters

Customers without dependents churn 5x more at 32.6%, versus 6.5% with dependents. Having a partner shows the same pattern but the difference is less extreme.

Age has some effect

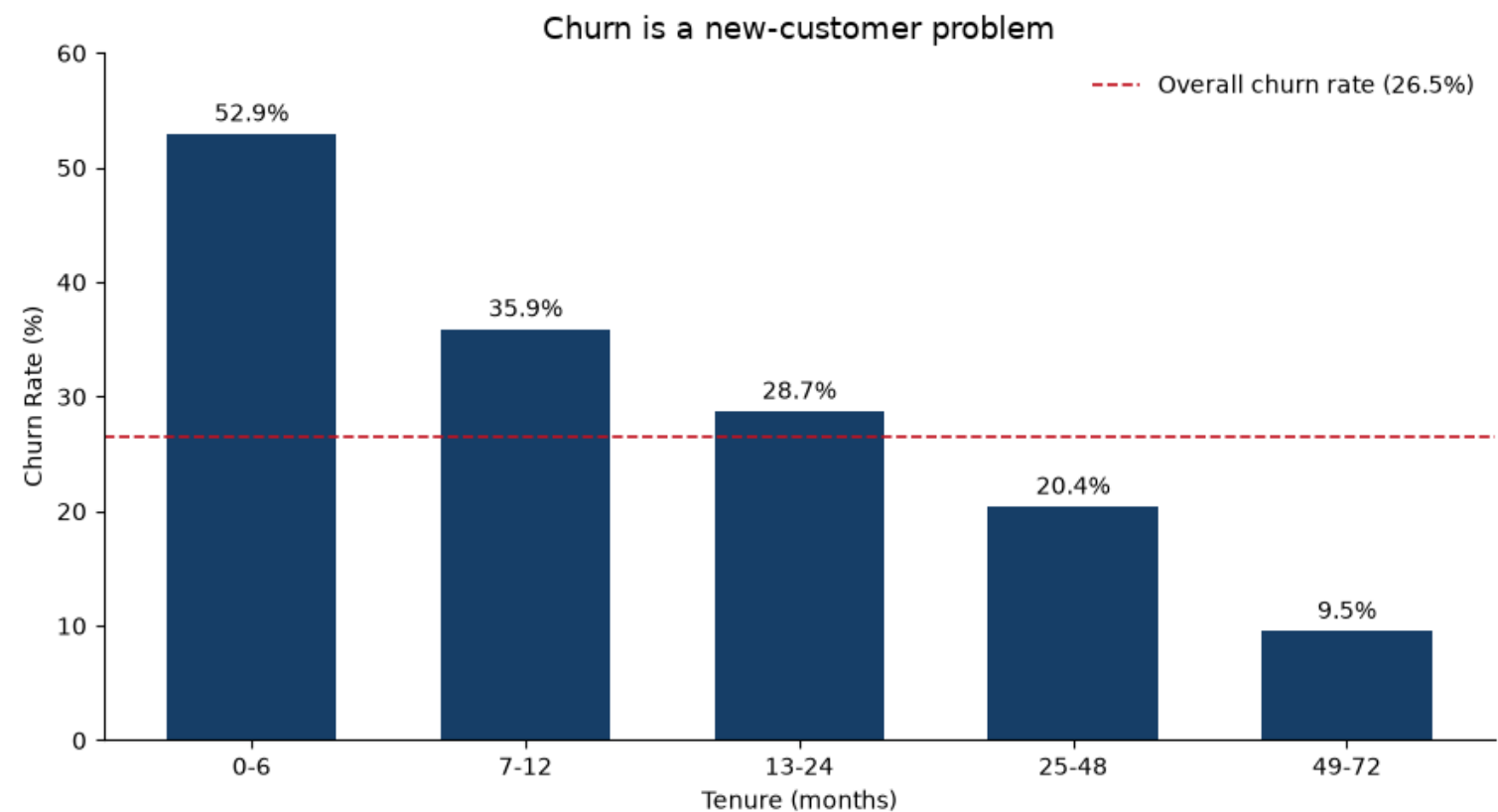
Senior citizens churn at 41.7%, nearly double the 23.6% rate for non-seniors.

Gender is not a driver

Churn is effectively flat between men (26.2%) and women (26.9%).

How Long They Stay

Churn is a new-customer problem



52.9%

of customers churn within their first six months, more than double the 26.5% average.

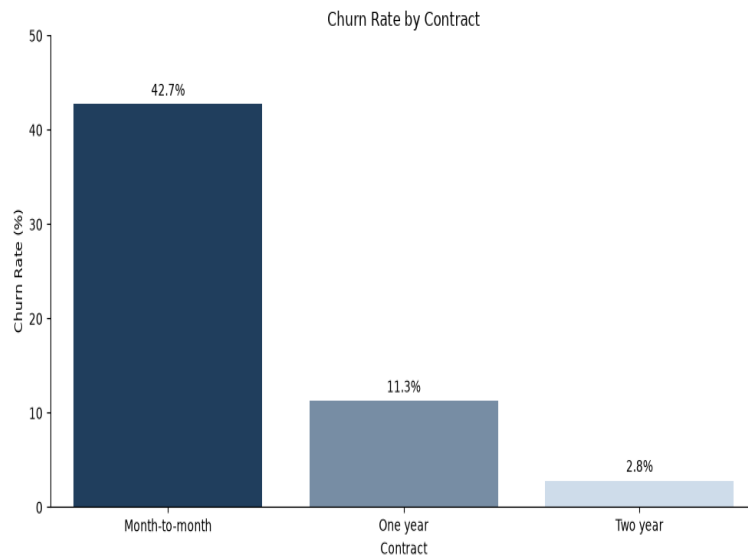
9.5%

is the churn rate for a customer between four and six years; tenure and loyalty go hand-in-hand.

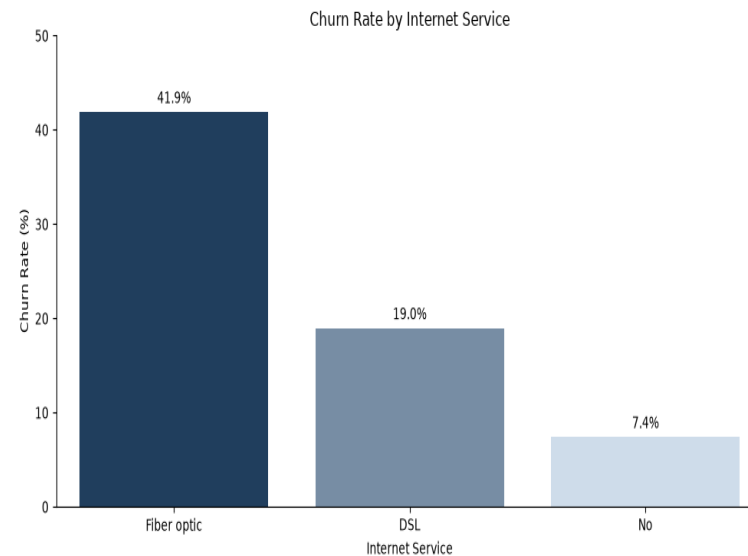
Implication: onboarding and the first six months are the highest-leverage window for retention effort.

What They Have

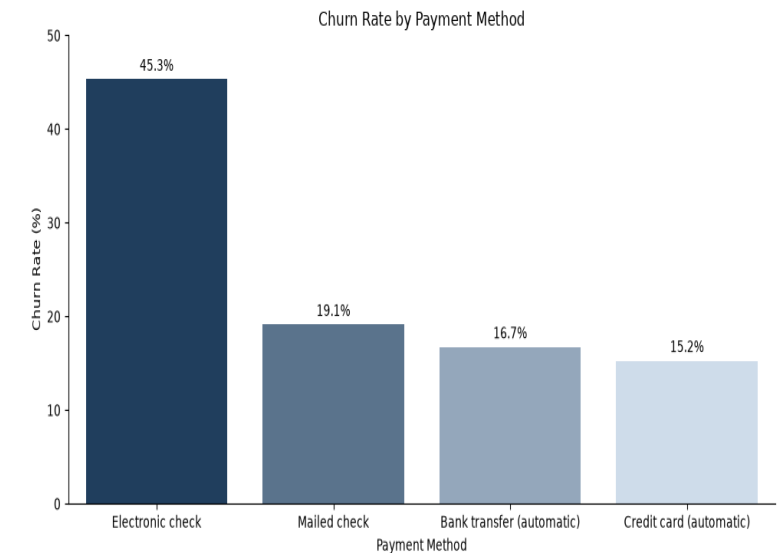
Contract, connection, and payment method



Month-to-month customers churn at 42.7%, versus 2.8% on a two-year contract.



Fiber customers churn at 41.9% – more than double DSL's 19.0%.

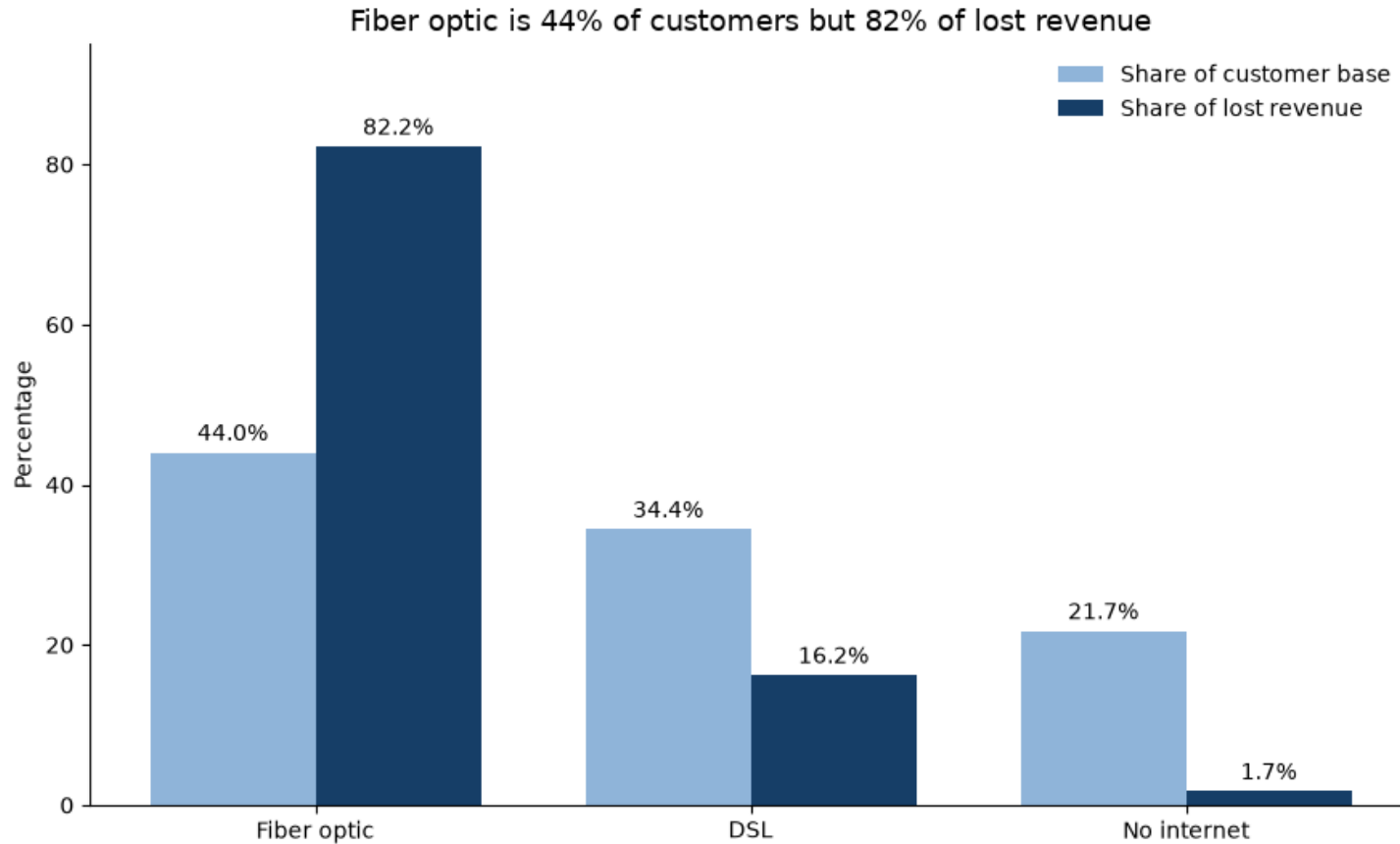


Electronic check payers churn at 45.3%, roughly 2.5x every other payment method.

These choices are associated with customer behaviours, making them potential pathways for improving customer loyalty.

The Fiber Problem

44% of customers, 82% of lost revenue



\$114,300

lost every month from churned fiber customers alone – \$1.37m across the year.

41.9%

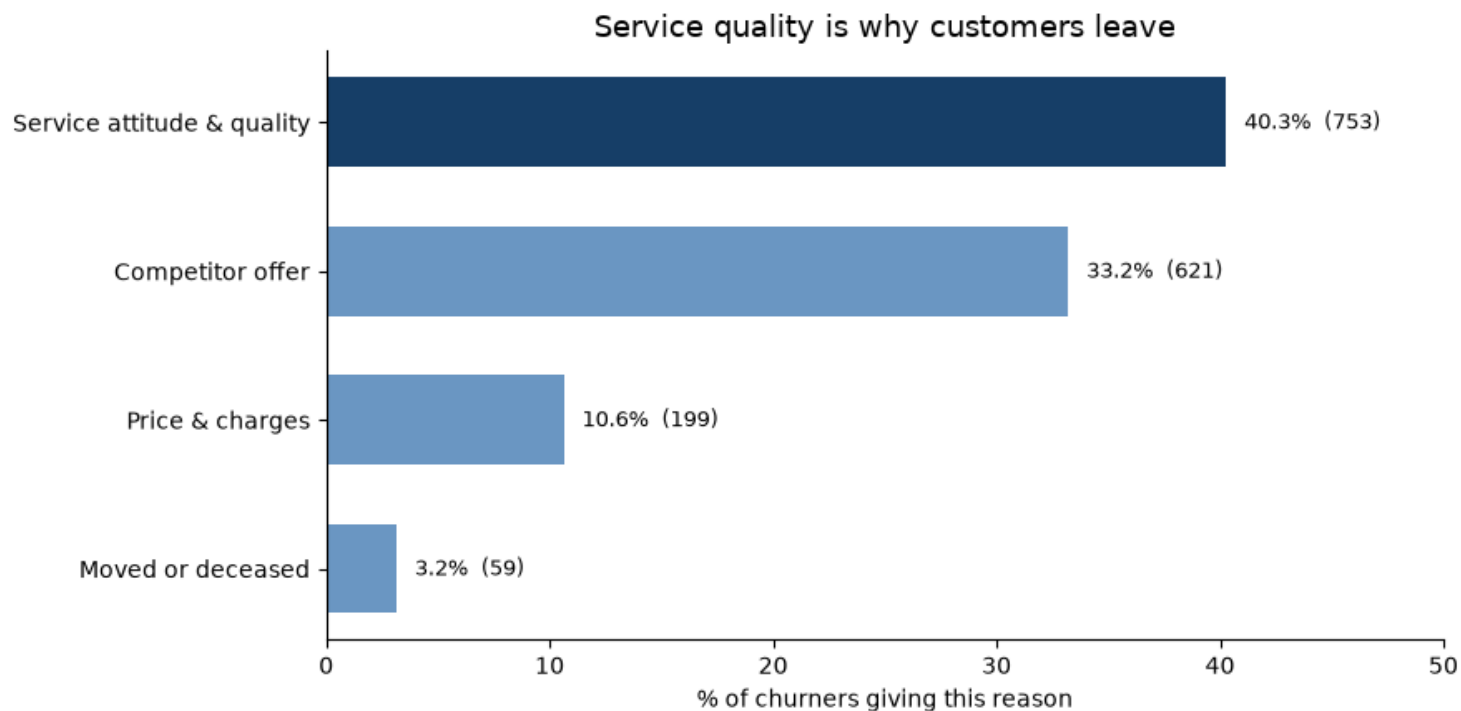
fiber churn rate, versus 19.0% for DSL and 7.4% for customers with no internet service.

2.2x

higher churn on fiber than DSL, making it the biggest revenue-at-risk segment.

Why They Leave

In their own words, from 1,869 exit reasons



Service quality: 40.3%

Attitude of support staff and support network dissatisfaction are the single largest theme.

Competitors: 33.2%

This could be better offers, more data, and higher download speeds elsewhere.

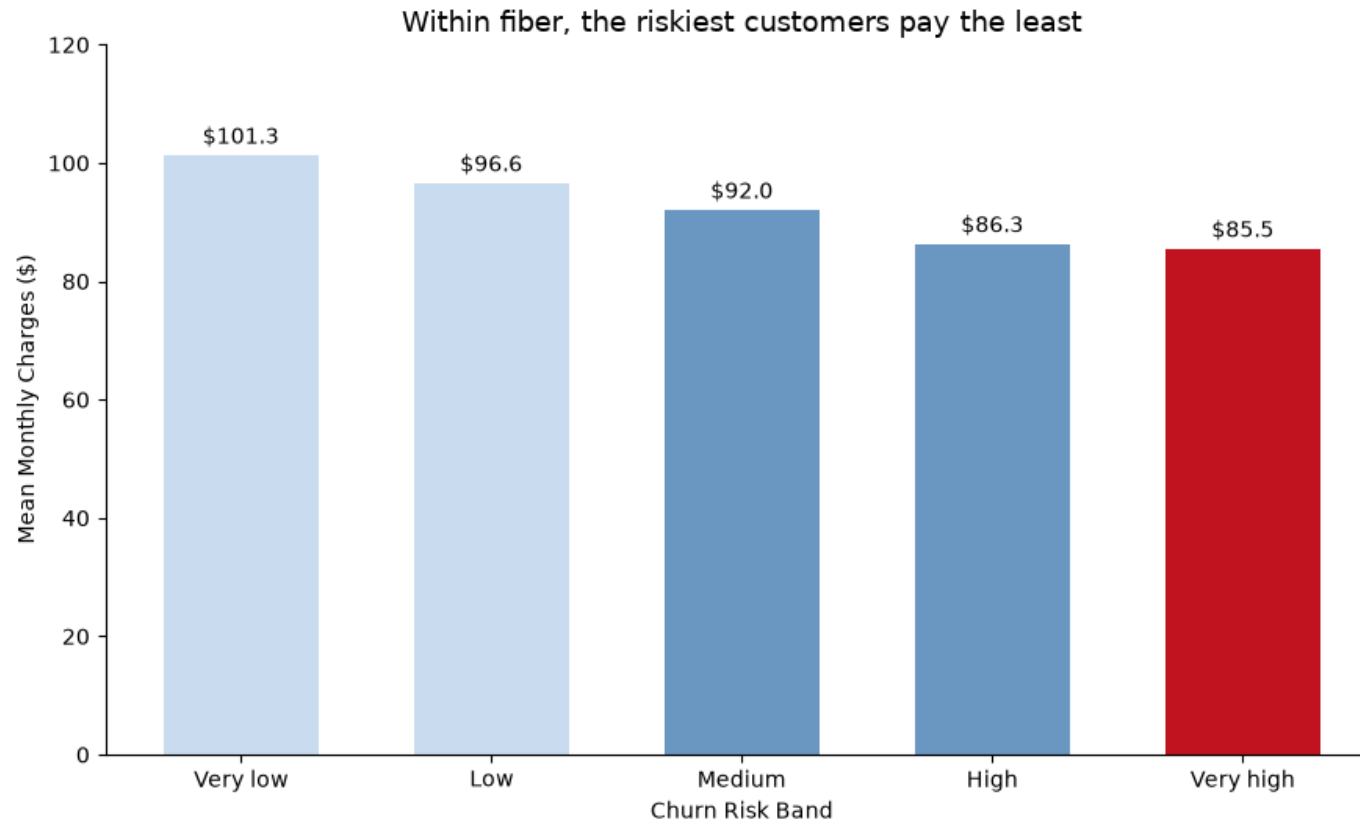
Price: 10.6%

Price is a factor worth considering, but not the dominant one.

The top two individual reasons, by name, are attitude of support staff and competitor offerings.

It Isn't Price

Low-risk customers pay more, why is that?



\$101 → \$85

average monthly fiber bill falls as churn risk climbs from Very low to Very high.

If price were the driver, our highest-risk fiber customers would be paying the most, not the least. This points to service experience, not cost, as the reason fiber customers leave.

What Drives Churn

Adjusted effect of each factor, all else held equal (logistic regression odds ratios)

2.37x

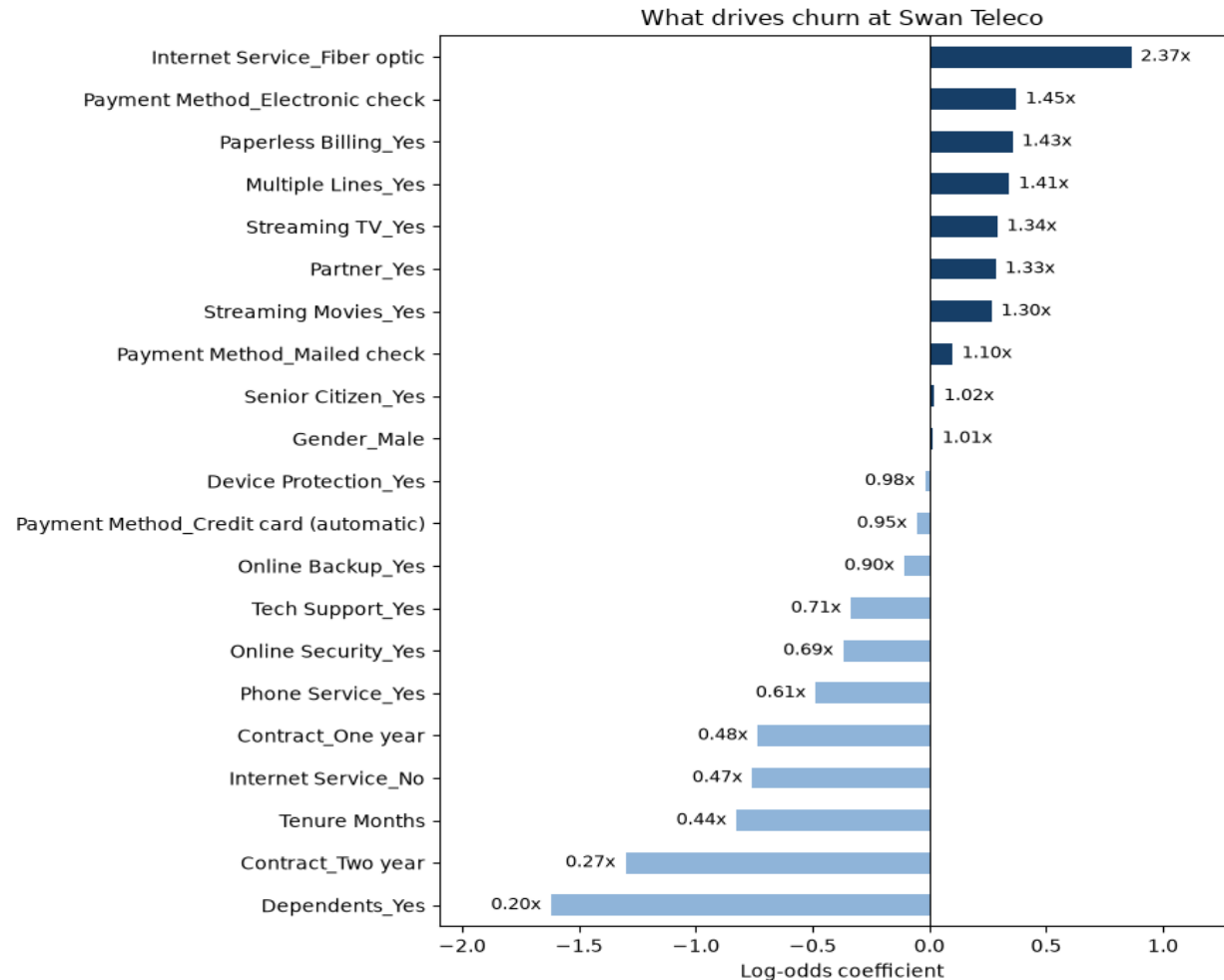
Fiber internet is the largest risk factor in raising churn odds.

0.20x

Having dependents cuts churn odds, making it a protective factor for retention.

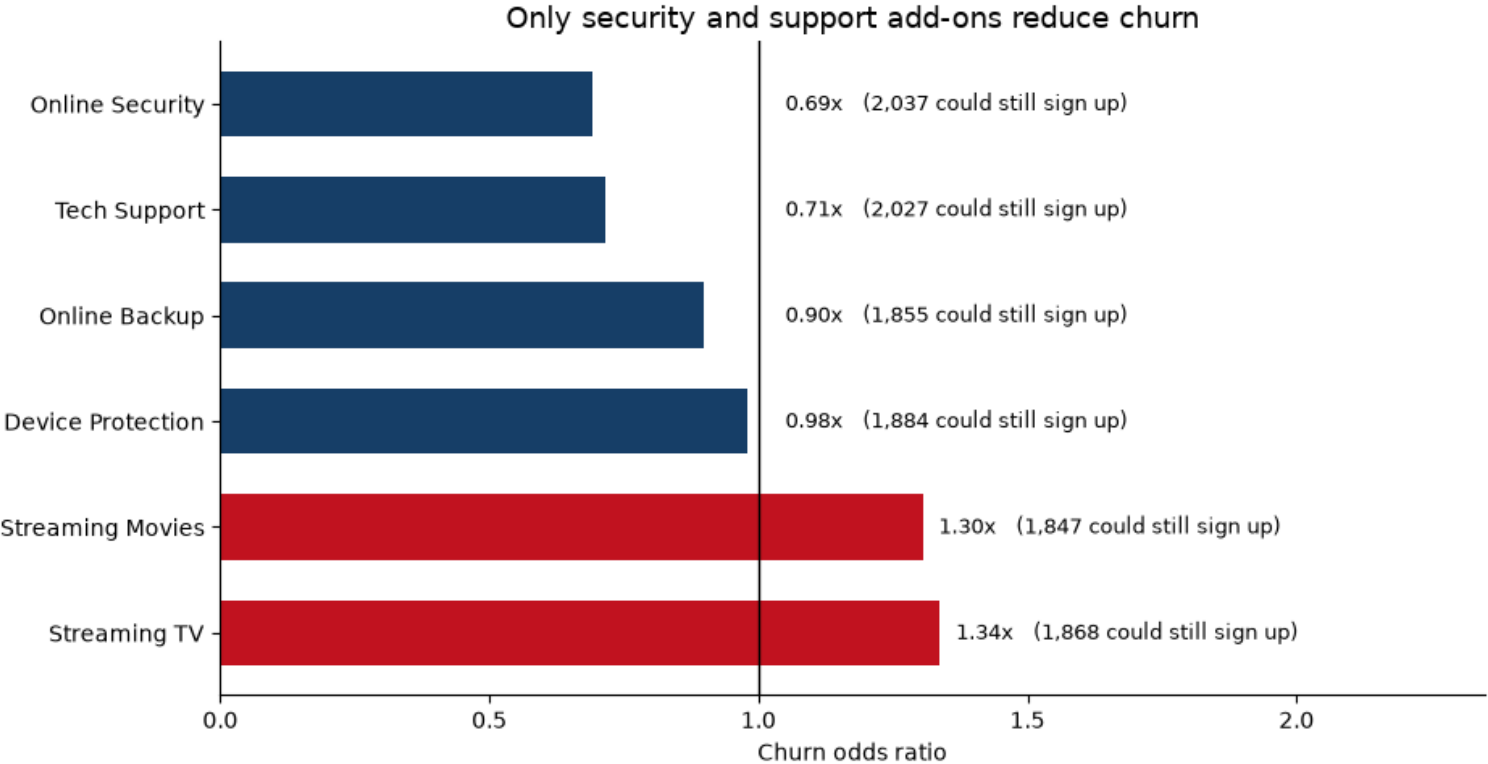
0.27x

A two-year contract lowers churn odds by nearly 75%.



Incentivise Online Security

Where the \$2.50 should go, and where it should not



0.69x

churn odds ratio for Online Security. Churn rate falls from 31.3% to 14.6% among customers who sign up. 2,037 retained customers don't have it yet.

0.71x

Tech Support carries a near-identical protective effect and a similar 2,027 customer headroom, making it a strong second incentive if budget allows.

Avoid incentivising Streaming TV or Streaming Movies as both raise churn odds (1.34x and 1.30x).

The 500 Mailers

64% mean churn risk against a 26.5% base rate

The 500 highest-risk retained customers were selected by ranking every retained customer on model churn probability. This group brings in \$40,038 of monthly revenue, and at a 64% mean churn risk we would expect 322 of them to leave. At the retention team's 20% offer uptake rate, the campaign should protect around 64 customers and \$5,188 a month, or about \$62,256 a year.

500

customers on the list

64%

mean churn risk (vs. 26.5% base)

\$40,038

monthly revenue represented

~64

customers expected to be saved

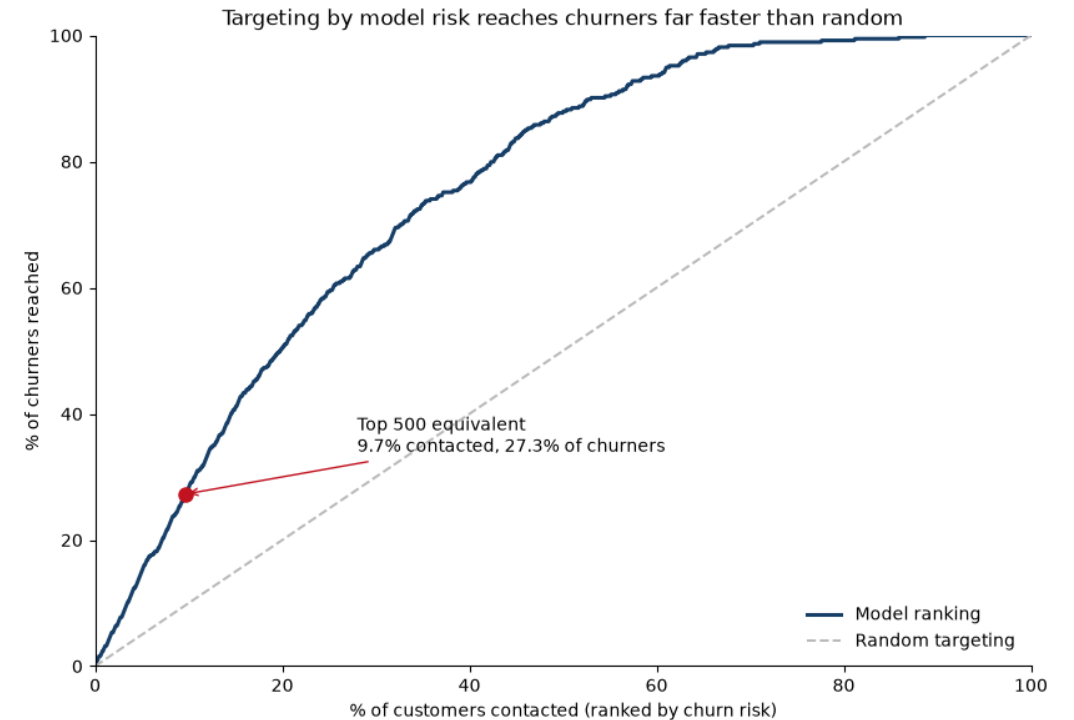
\$5,188

monthly revenue protected

\$62,256

protected annually

The full ranked list of these 500 customers is provided as a companion CSV file.

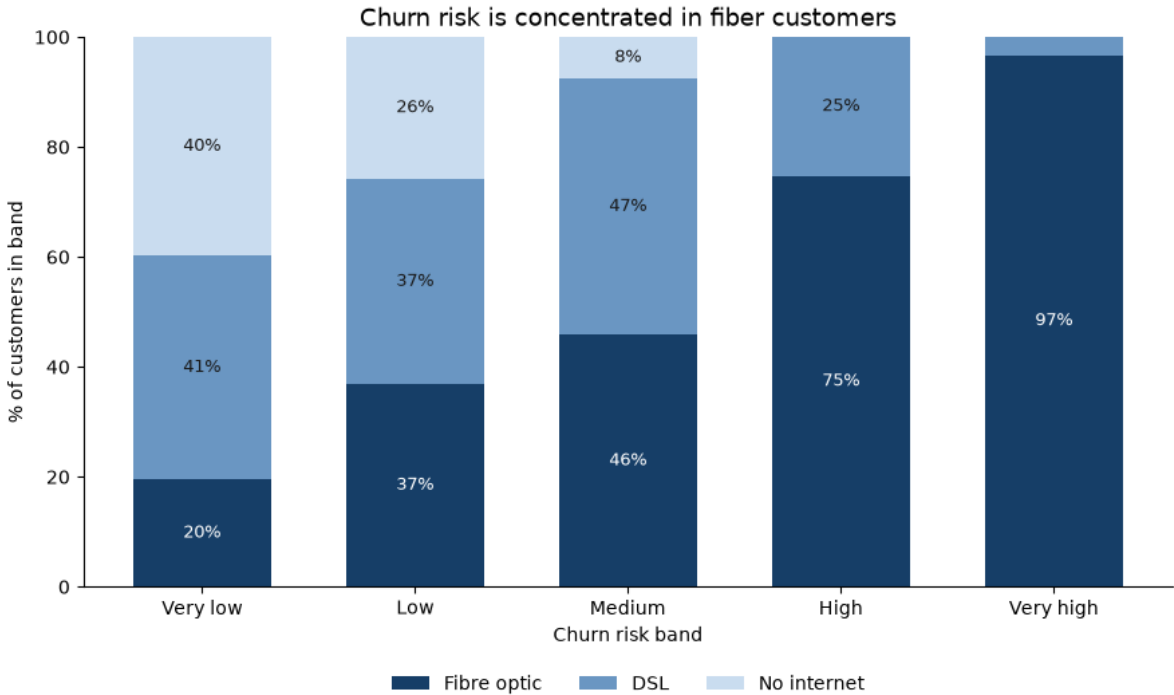


Churn Risk Register

A banded risk score for every remaining customer

All 5,174 retained customers are scored and banded from Very low to Very high risk. The problem concentrates in 665 people: 215 Very high (96.7% of them on fiber) and 450 High.

Risk band	Customers	Share	Mean risk	Monthly revenue
Very low	2,725	52.7%	3.4%	\$150,048
Low	1,052	20.3%	16.6%	\$65,009
Medium	732	14.1%	33.8%	\$49,150
High	450	8.7%	54.6%	\$34,756
Very high	215	4.2%	72.4%	\$18,024



Every remaining customer's individual band and score is provided as a companion CSV file for customer service.

What We Recommend

Three actions to consider, in order of value

1

Run the 500-customer mailer campaign

Target the highest-risk retained customers now. Expected to protect ~\$62,000 a year at a 20% uptake rate.

2

Put the \$2.50 sign-up incentive on Online Security

It carries the strongest protective effect (0.69x odds) and 2,037 customers still don't have it. Avoid incentivising streaming add-ons, which raise churn odds.

3

Commission a fiber optic product and service review

Fiber is 44% of customers but 82% of lost revenue, and churns at 2.2x DSL while its own riskiest customers pay the least.

Limitations

Insights built off a snapshot, not a forecast

The model is trained on 3 months of data. Customer behaviour, competitor offers, and pricing will shift, and the model should be refreshed regularly.

Moderate recall on churners

The model catches 57% of true churners at the default threshold (80% overall accuracy, 0.85 AUC). This is useful for prioritisation, but it will still miss some at-risk customers.

Association, not proven cause

The model shows which factors go together with churn. It doesn't prove that changing one will make churn fall by that amount. Signing customers up to Online Security should help, but the only way to know how much is to trial it on part of the base first.

Churn reasons may not be a true reflection of a customer's feelings

Exit reasons reflect what customers chose to say, which may not capture the full or true story behind their decision to leave.